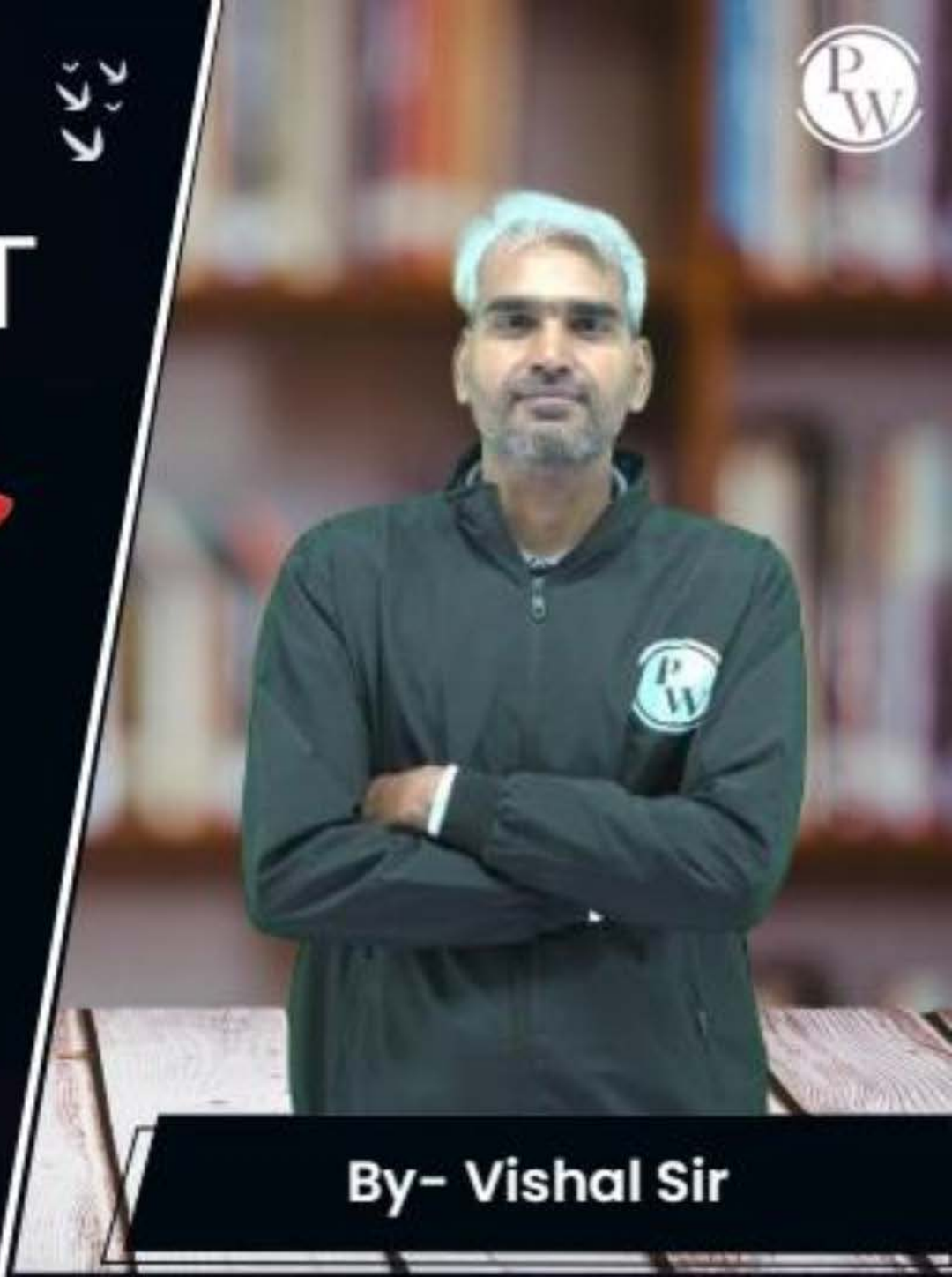
Computer Science & IT



Compiler Design

Syntax Directed Translation

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- **Topic** — SDD and SDT
- **Topic** — Types of attributes
  - Synthesized
  - Inherited
- **Topic** — S-attributed and L-attributed SDT
- **Topic** — Questions based on S-attributed SDT

Slide

Topics to be Covered



- **Topic** Difference between SDD and SDT
- **Topic** Syntax-Directed Translation Schemes for S-attributed SDD
- **Topic** Syntax-Directed Translation Schemes for L-attributed SDD
- **Topic** Questions based on L-attributed SDT



Topic : SDD vs SDT

SDD

- CFG along with attributes and semantic rules, but we do not define the order in which semantic rules will be evaluated

SDT

Translation scheme OR SDT will define the order in which semantic rules/actions are performed

Slide



Topic : S-attributed SDD

- A syntax directed definition in which all attributes are synthesized attribute

eg:

$E \rightarrow E_1 + T$	$\{E.val = E_1.val + T.val\}$
$E \rightarrow T$	$\{E.val = T.val\}$
$T \rightarrow T_1 * F$	$\{T.val = T_1.val * F.val\}$
$T \rightarrow F$	$\{T.val = F.val\}$
$F \rightarrow id$	$\{F.val = id.lexval\}$

- All attributes are synthesized,
  ∴ S-attributed SDD

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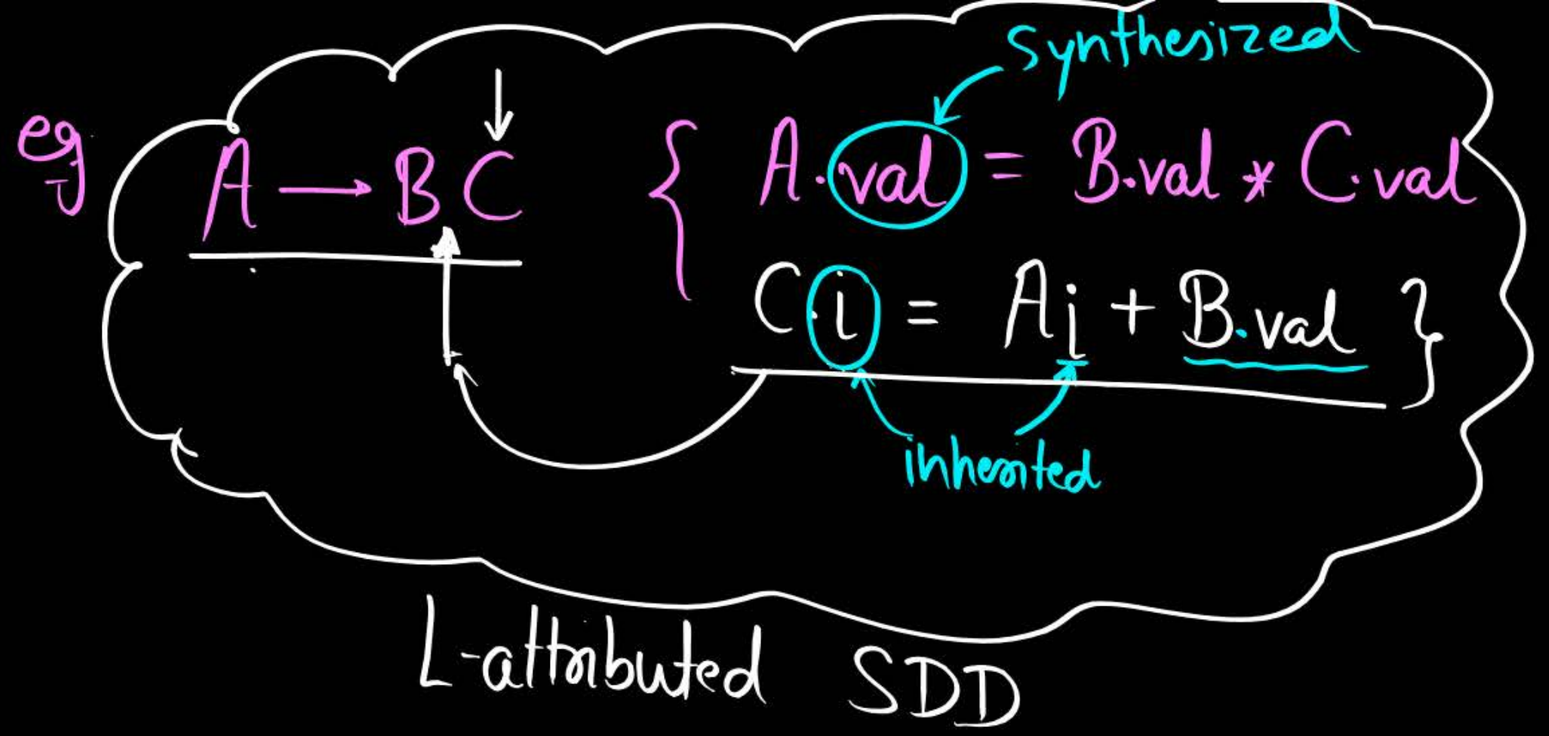
Topic : L-attributed SDD

L-attributed SDD Can have

① Synthesized attributes

and/or ② Inherited attributes with the restriction that values Can be inherited from the inherited attributes of Parent node and/or inherited/Synthesized attributes of left siblings.

Slide


eg
A → B C
{ A.val = B.val * C.val
C.i = Ai + B.val }
Synthesized
inherited
L-attributed SDD
'i' is the inherited attribute for non-terminals C and A

$A \longrightarrow B\ C\ D\ E$ { let i = inherited attribute, s = Synthesized attribute }

$D.i = f(A.s, A.i, B.s, B.i, C.s, C.i, E.s, E.i)$

inherited attribute

$A.s = g(B.i, B.s, C.i, C.s, D.i, D.s, E.i, E.s)$

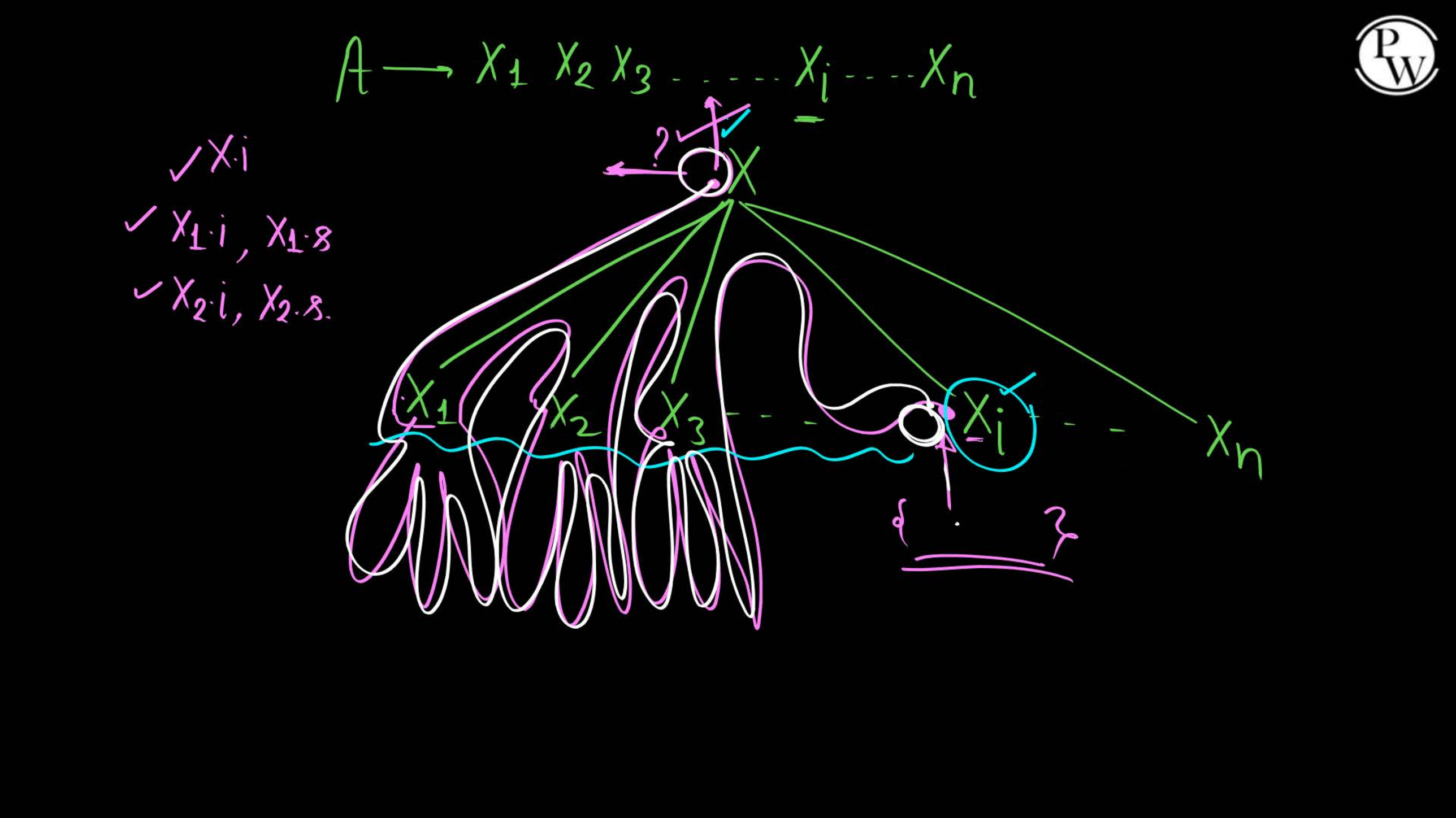


Topic : NOTE

Syntax-Directed Translation Schemes for S-attributed SDD

For S-attributed SDD we can construct an SDT in which each semantic action/rule is placed at the end of the production and that semantic action/rule is executed whenever reduce action is performed by bottom-up parser corresponding to that production.

SDT's with all actions at the right ends of the production bodies are called postfix SDT's.


A → X1 X2 X3 ..... Xi ---- Xn
✓ X.i
✓ X1.i , X1.8
✓ X2.i , X2.8.
?
X
X1
X2
X3
Xi
Xn



Topic : NOTE





Syntax-Directed Translation Schemes for L-attributed SDD

The rules for turning an L-attributed SDD into an SDT :

1. Embed the action that computes the inherited attributes for a nonterminal A immediately before that occurrence of A in the body of the production. If several inherited attributes for A depend on one another in an acyclic fashion, order the evaluation of attributes so that those needed first are computed first.
2. Place the actions that compute a synthesized attribute for the head(non-terminal) of a production at the end of the body of that production.

Slide



Topic : Evaluation strategy for S-attributed SDT

While parsing the given input string in bottom-up manner, whenever reduce action is performed, semantic rule/action associated with that production is evaluated.



Topic : Evaluation strategy for L-attributed SDT

- Construct the parse tree for the given input string.
- While constructing the parse tree create nodes corresponding to the semantic rules/actions as well in the parse tree. (for differentiation purpose connect those nodes with the parent node using dotted lines).
- Traverse that parse tree in depth-first manner from left to right. Whenever node associated with a semantic rule/action is visited corresponding rule/action is performed.
  - Every S-attributed SDT is also L-attributed SDT, Therefore same strategy can always be applied on S-attributed SDT.



Topic : NOTE

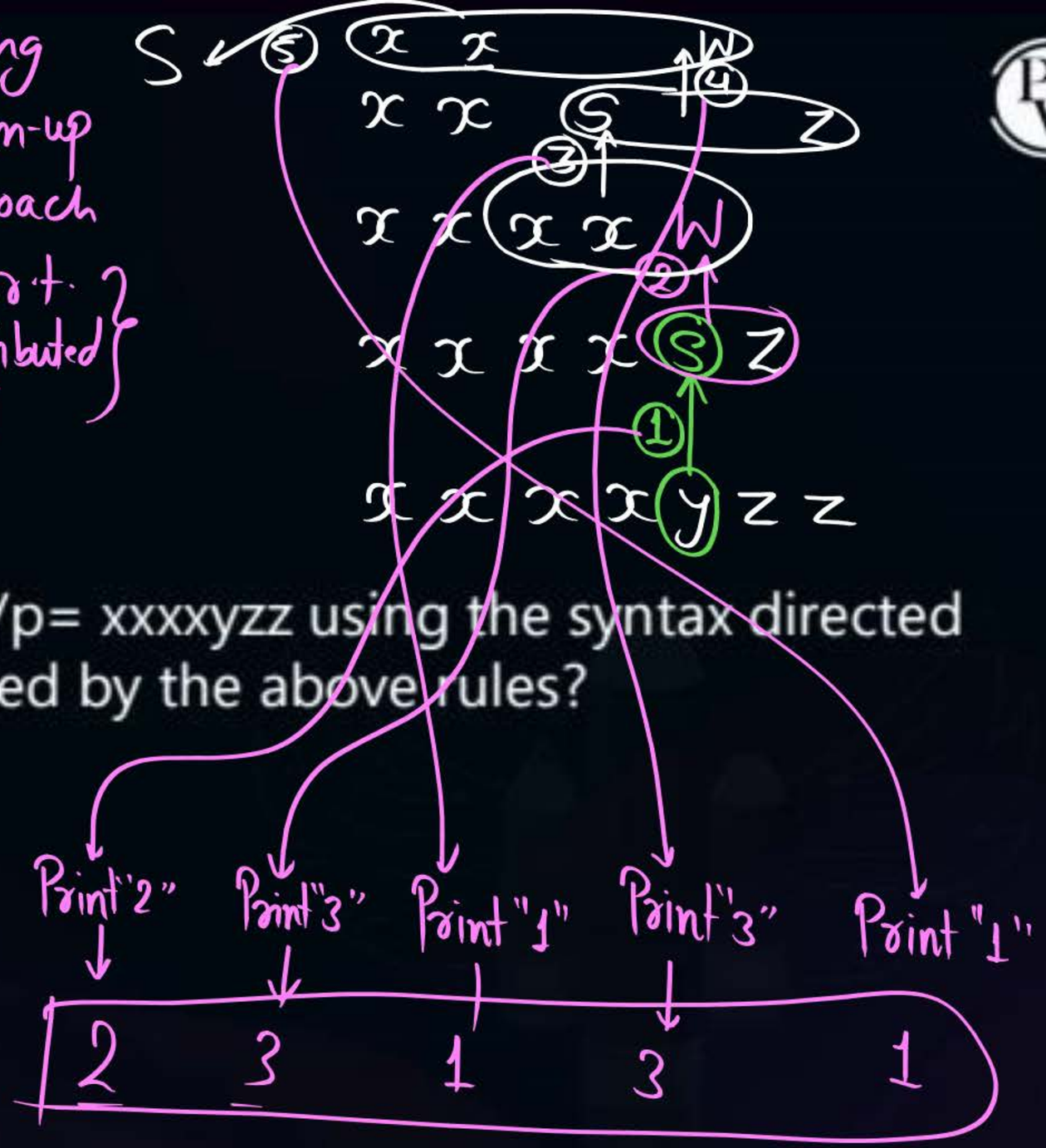
Using Bottom-up Approach { i.e. w.r.t. S-attributed SDT }

Q. S → xxW { print"1"}

S → y { print"2"}

W → Sz { print"3"}

What is the translation of i/p= xxxxyzz using the syntax directed translation scheme described by the above rules?



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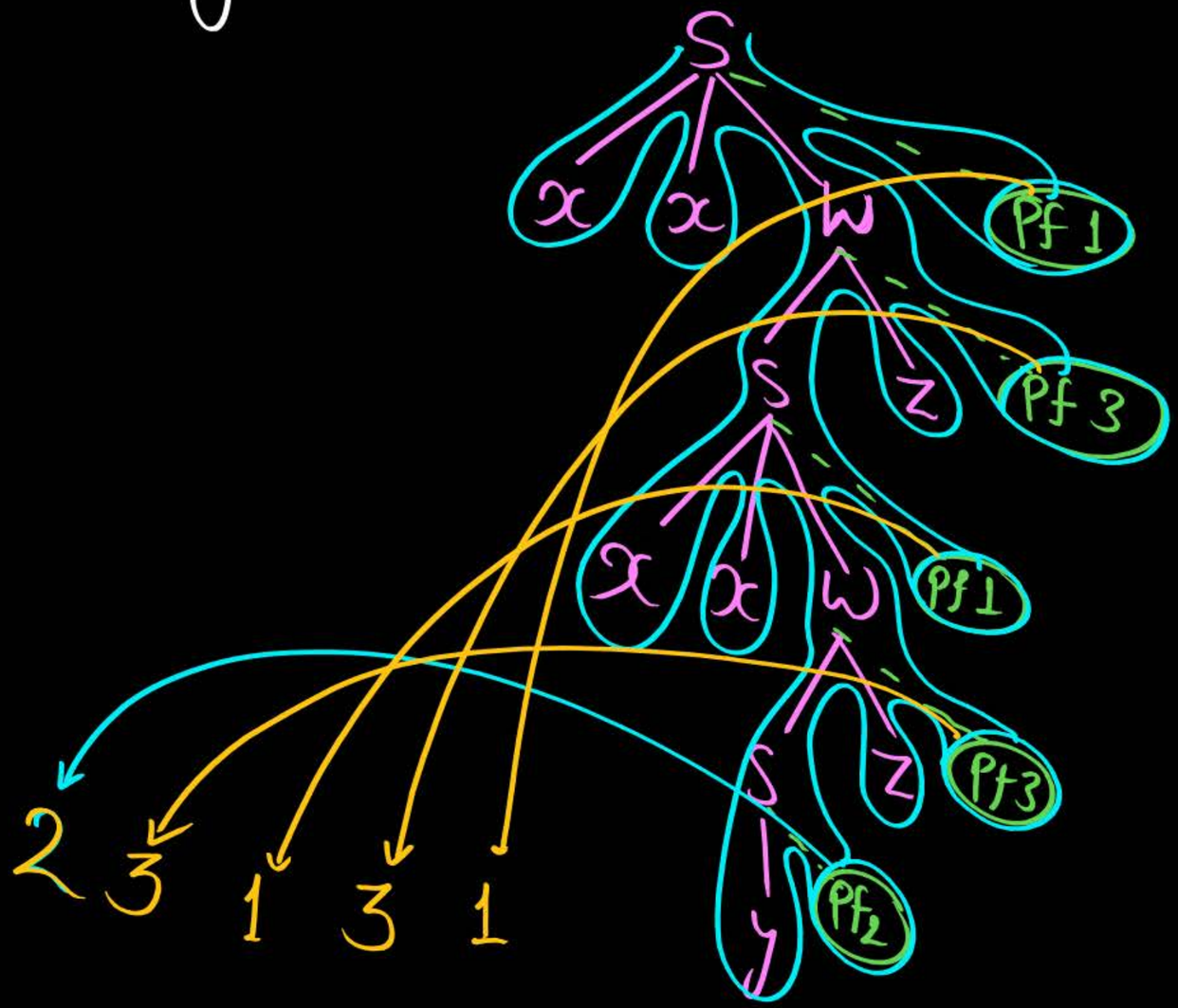
Above SDT is S-attributed as well as L-attributed

∴ it can be translated using top-down approach as well

① Construct the parse tree

② Create node w.r.t Semantic rule/action

③ Traverse tree depth-first manner (left to Right)



Topic : Evaluation of L-attributed SDT

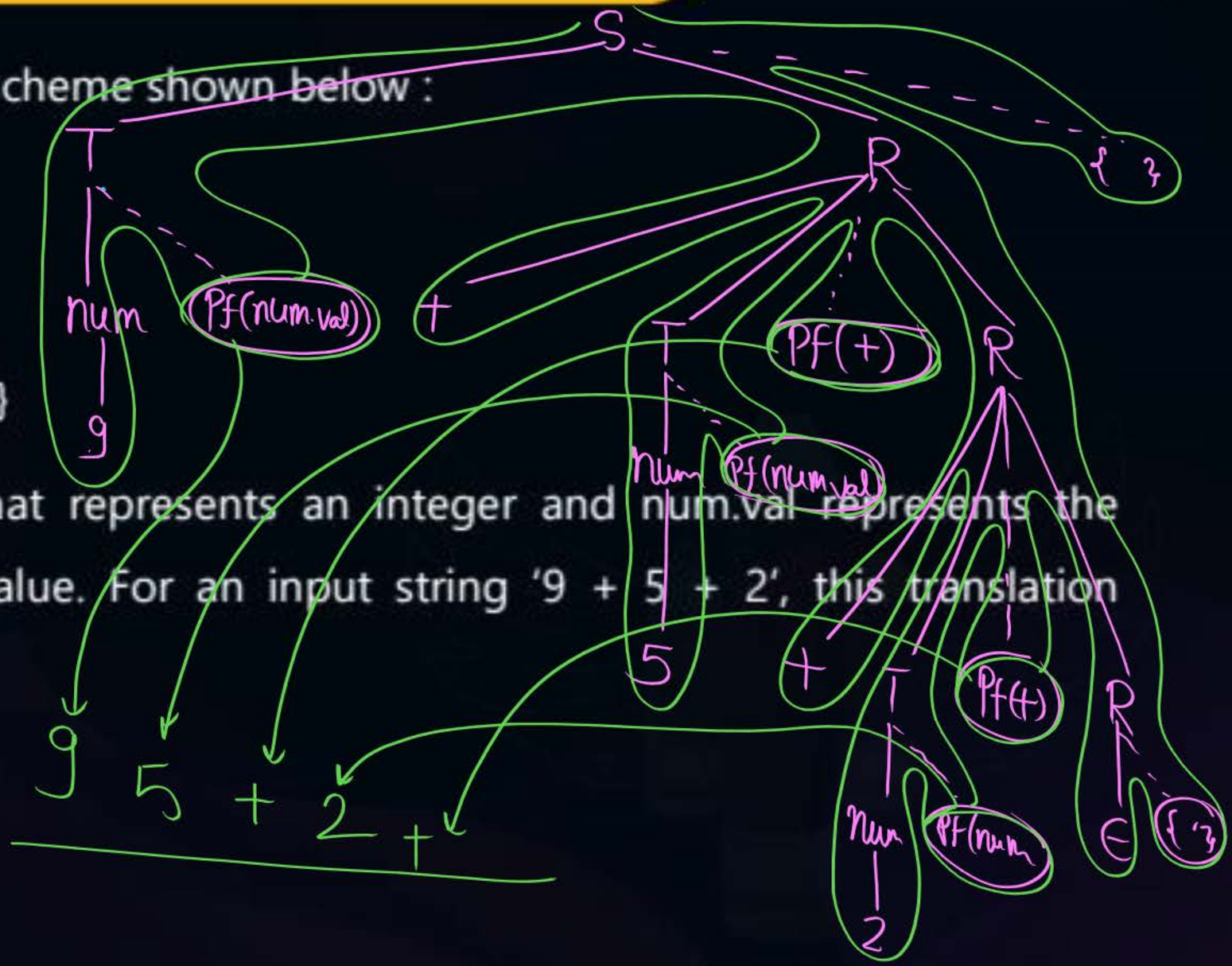
Consider the translation scheme shown below :

$S \rightarrow TR$

$R \rightarrow +T$ {print ('+');} $R|\epsilon$

$T \rightarrow$ num {print(num. val);}

Here num is a token that represents an integer and num.val represents the corresponding integer value. For an input string '9 + 5 + 2', this translation scheme will print



Topic : Evaluation of L-attributed SDT

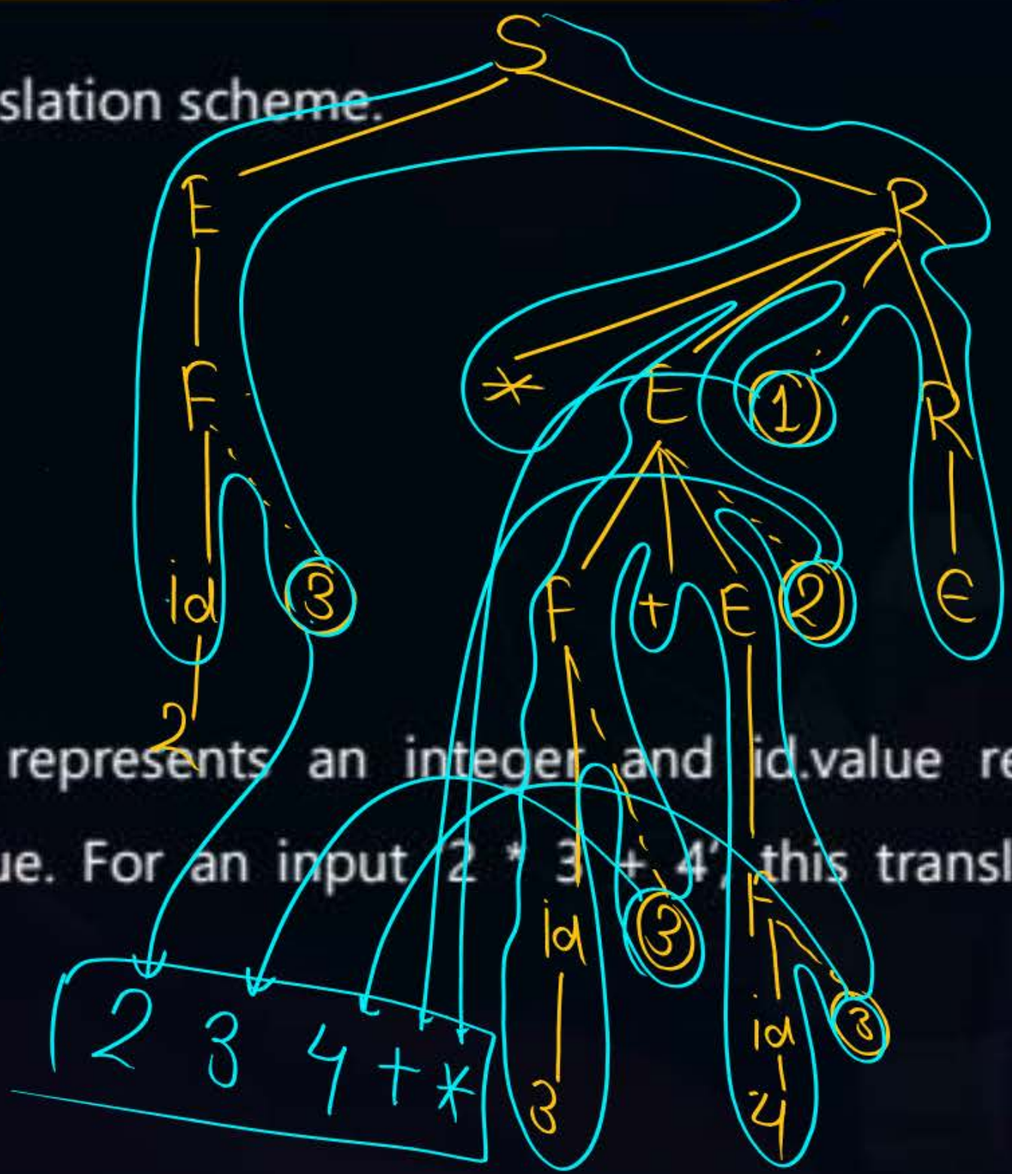
Consider the following translation scheme.

$S \rightarrow ER$

$R \rightarrow *E\{print(\text{“*”});\}\ R \mid \epsilon$

$E \rightarrow F + E\ \{print(\text{“+”});\} \mid F$

$F \rightarrow (S) \mid id\ \{print(id.value);\}$



Here id is a token that represents an integer and id.value represents the corresponding integer value. For an input '2 * 3 + 4', this translation scheme prints .....?

2 3 4 + *

Slide



Topic : Application of Syntax Directed Translation

SDT can be used for:

1. Evaluating given arithmetic expression
2. Infix to postfix expression
3. Conversion from Binary to Decimal
4. Generating intermediate code
5. Type Checking

And many more



Topic : SDT for Evaluating arithmetic expression

$E \rightarrow E + T$	{E. Val = E.Val + T. Val}
$E \rightarrow T$	{E.Val = T.Val}
$T \rightarrow T*F$	{T. Value = T. value * F. Val}
$T \rightarrow F$	{T. Value = F. Val}
$F \rightarrow num$	{F. Val = num.Val}

Already done

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Topic : SDT for Infix to postfix expression

$2+3*4 \equiv 2\ 34*\ +$

1

$E \rightarrow E + T$ {print ("+")}

$E \rightarrow T$ {}

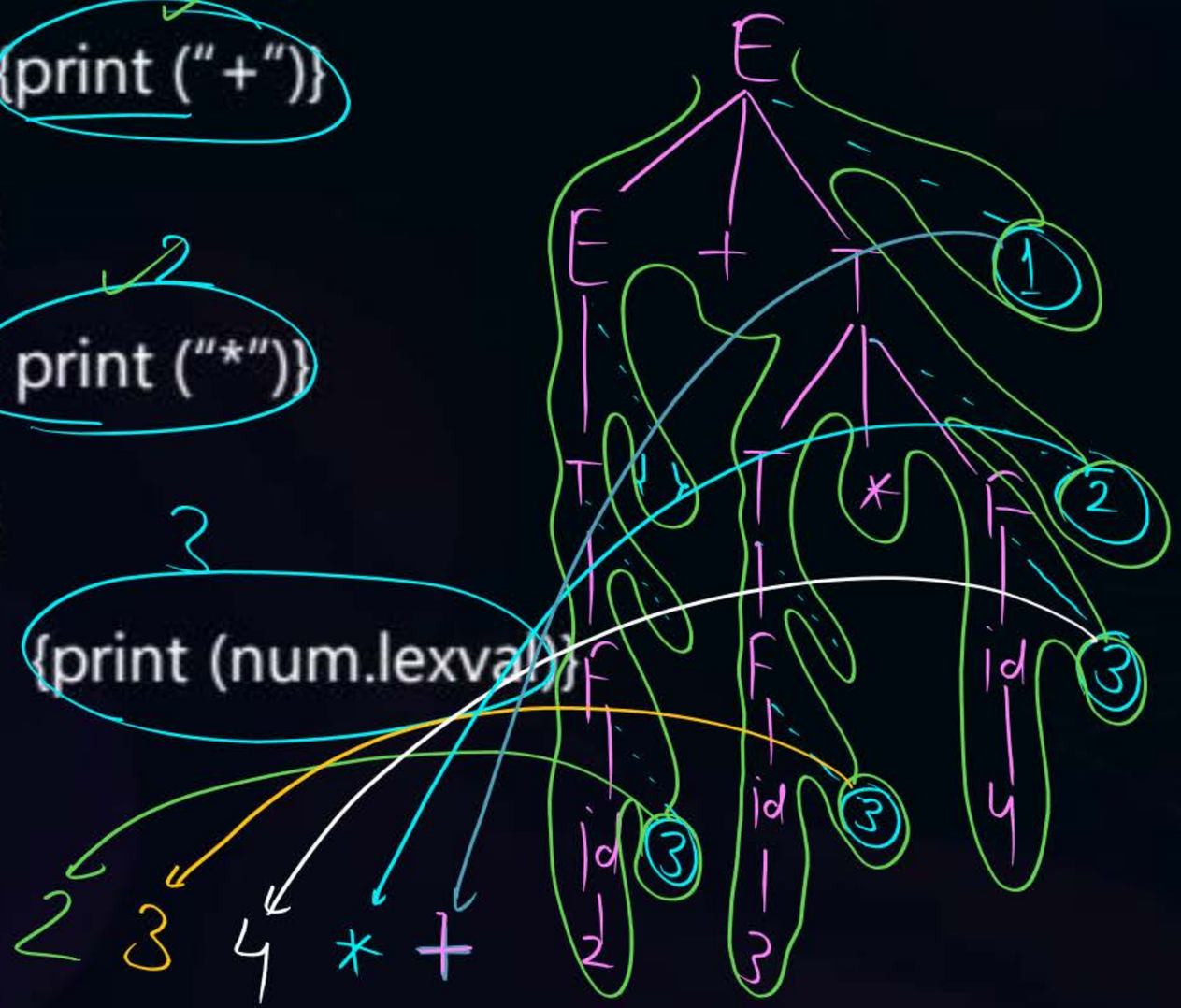
2

$T \rightarrow T * F$ { print ("*")}

$T \rightarrow F$ {}

3

$F \rightarrow num$ {print (num.lexval)}



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Topic : SDT for Binary to Decimal Without fraction

Number list of bits

$N \rightarrow L \{N.dval = L.dval\}$

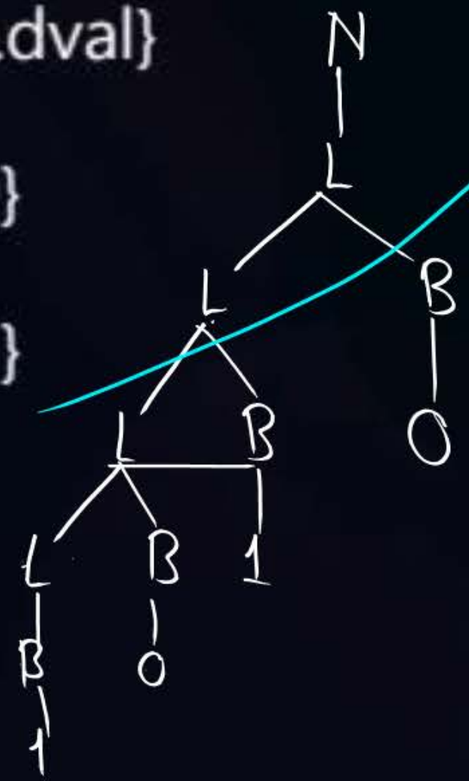
$L \rightarrow L_1 B \{L.dval = L_1.dval * 2 + B.dval\}$

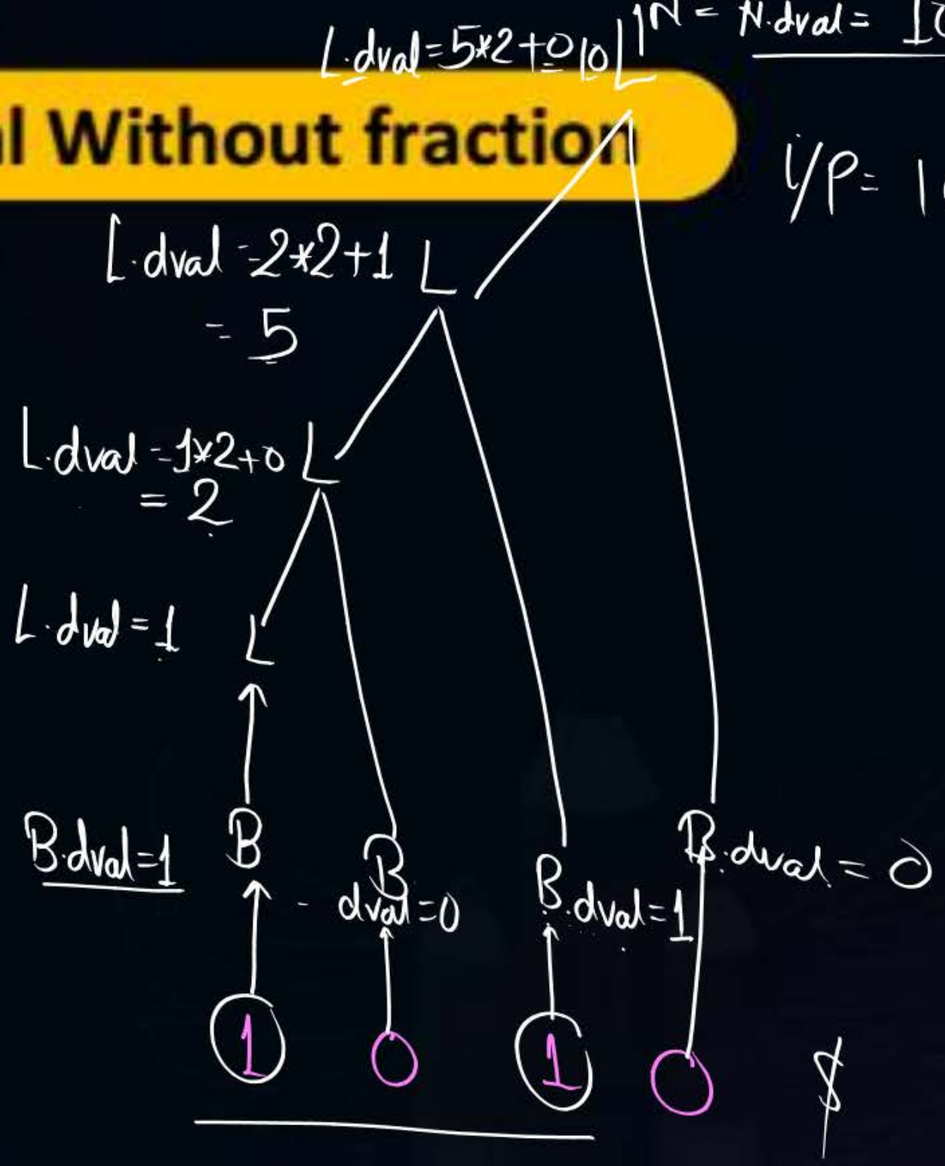
(B: bit)

$L \rightarrow B \{L.dval = B.dval\}$

$B \rightarrow 0 \{B.dval = 0\}$

$B \rightarrow 1 \{B.dval = 1\}$





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Topic : SDT for Code generation

Consider the syntax directed definition shown below.

```
S → id : = E   {gen (id.place = E.place;);}
E → E1 + E2    {t = newtemp ( ); gen (t = E1.place + E2.place;); E.place = t}
E → id         {E.place = id.place;}
```

Here, gen is a function that generates the output code, and newtemp is a function that returns the name of a new temporary variable on every call. Assume that ti's are the temporary variable names generated by newtemp. For the statement 'X: = Y + Z', the 3-address code sequence generated by this definition is

(A) X = Y + Z

(B) t1 = Y + Z; X = t1

(C) t1 =Y; t2 = t1 + Z; X = t2

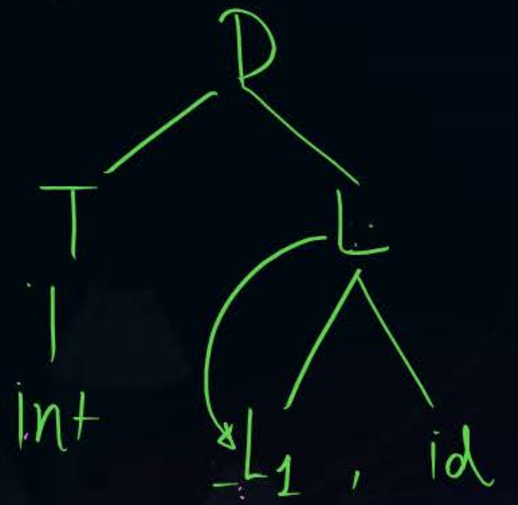
(D) t1 = Y; t2 = Z; t3 = t1 + t2; X = t3

Slide

Topic : Question

Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let X1, X2, X3. X4, X5, and X6 be the placeholders for the non-terminals D, T, L or L1 in the following table:

Production rule	Semantic action
$D \rightarrow TL$	$X_1.type = X_2.type$
$T \rightarrow int$	T.type = int
$T \rightarrow float$	T.type = float
$L \rightarrow L_1, id$	$X_2.type = X_4.type$ add Type (id.entry, X_5.type)
$L \rightarrow id$	add Type (id.entry, X_6.type)



Which one of the following are the appropriate choices for X_1, X_2, X_3 and X_4?

Slide

A $X1 = L, X2 = T, X3 = L1, X4 = L$

B $X1 = T, X2 = L, X3 = L1, X4 = T$

C $X1 = L, X2 = L, X3 = L1, X4 = T$

D $X1 = T, X2 = L, X3 = T, X4 = L1$

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Topic : NOTE

① S-attributed SDT :-

* Grammar is LR Parsable and definition is S-attributed

② L-attributed SDT :-

* Grammar is LL parsable, and definition is L-attributed

Slide



Topic : Attribute grammar

* Attribute grammer is SDD in which semantic rules have no side effect.
  Side effect can be any fragment of code.
* { Attribute grammar is a CFG along with attribute, and semantic rules free from side effect }
* In Attribute grammar only the values of attributes is evaluated

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2 mins Summary

Topic SDD vs SDT

Topic Types of SDT and their evaluation

Slide

THANK - YOU